Gabriel Vance
DSC640 – Data Presentation & Visualization
Professor Benjamin Schneider
September 14, 2025

Milestone 1

Initial Impressions:

The National Database of Childcare Prices maintains an exhaustive list of data that goes beyond the cost of childcare. While that might be the sole purpose for the data, one glimpse and we can find other branching topics, such as unemployment rate, stratified by age and gender. Each row corresponds to a time and a place, such as Autauga County, Alabama in 2009. The primary key will likely be the State, Count_Name, and StudyYear. The various columns provide critical information, with a mix of categorical codes and numerical data.

The volume of data is more than we need for a simple analysis, as the data contains 227 columns. However, 15 columns have the same values across all rows, so they can likely be dropped. If one were to use this data source for a study, they would have to decide whether they want to do a deep dive into a specific demographic, such as females between 20 and 64 with children under 6, or the general population. The large number of columns leave the possibility to pivot into different areas.

There appear to be various “target” columns, such as Median Price for Centers, Median Price for Family Child Care, 75th Percentile Price for Centers, 75th Percentile Price for Family Child Care. For our purposes, we’ll focus on Median Price for Centers. We will explore the rising costs overall and for each unique age group (such as birth to 5 months, 6 months to 11 months, etc.

There are also 2 forms of incomes/earnings, one using the target year and one using 2018 dollars. 2018 dollars take inflation into account, which provides a better picture of the economy, so we’ll use those columns. While the data provides columns for households comprising two races, the majority of datapoints fall under 1 race, such as black, white, Asian, Hispanic. We will use the 1 race columns for our efforts, should we pursue insights into racial demographics.

Scenario:

For our efforts, we will use this data as the source for a political movement. We are imagining that the year is 2028 and the presidential election is upon us, and the key issue is childcare costs. Both candidates are neck-and-neck in the polls, and we have a last campaign push to win the election. Our goal is to present the data with a compelling

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argument that we understand the cost of childcare and what the driving factors are. Here are the questions we aim to answer:

1. What is the average cost of childcare?
2. What percent of income does the average household spend on childcare?
3. Is the data evenly distributed or is it skewed (indicating imbalanced in childcare costs)?
4. Do childcare costs cause poverty/unemployment?

Because we need to choose 3 different mediums, we will choose 3 different audiences.

1. Video – This will be a campaign speech. The duration will be between 3-4 minutes. This will provide us the opportunity to describe the problem, present a solution, and show off our expertise.
    a. The audience will be the **audience at the campaign speech** and **YouTube viewers** who will likely watch only the 3–4-minute snippet.
2. PowerPoint – This will be an internal "factoid" PowerPoint that will be up on the campaign website. It will be a brief set of slides (3-5) that rephrase the campaign's position on childcare costs.
    a. The audience of the PowerPoint will be **campaign volunteers** who will learn the facts so they can present them
3. Single page PDF printout – This will contain an even more condensed version of the PowerPoint. This will be for infographic leaflets to be handed out as physical copies.
    a. The audience of the leaflets will be voters on the streets, door-to-door voters,

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Purpose: This notebook performs the tasks outlined in Milestone 1. The full assignment will join the work found within this notebook with other milestones to create a final report.

In [1]:
```
import pandas as pd

df = pd.read_excel('nationaldatabaseofchildcareprices.xlsx')
```

In [2]:
```
# read the sample data
df.head()
```

Out[2]:

	State_Name	State_Abbreviation	County_Name	County_FIPS_Code	StudyYear	UN
0	Alabama	AL	Autauga County	1001	2008	
1	Alabama	AL	Autauga County	1001	2009	
2	Alabama	AL	Autauga County	1001	2010	
3	Alabama	AL	Autauga County	1001	2011	
4	Alabama	AL	Autauga County	1001	2012	

5 rows × 227 columns

In [3]:
```
# check the data types and columns
print(df.dtypes)
print(df.columns)
```

```
State_Name               object
State_Abbreviation       object
County_Name              object
County_FIPS_Code          int64
StudyYear                 int64
                         ...
_75FCCInfant_flag       float64
_75FCCToddler           float64
_75FCCToddler_flag      float64
_75FCCPreschool         float64
_75FCCPreschool_flag    float64
Length: 227, dtype: object
Index(['State_Name', 'State_Abbreviation', 'County_Name', 'County_FIPS_Cod
e',
       'StudyYear', 'UNR_16', 'FUNR_16', 'MUNR_16', 'UNR_20to64',
       'FUNR_20to64',
       ...
       'MFCCToddler', 'MFCCToddler_flag', 'MFCCPreschool',
       'MFCCPreschool_flag', '_75FCCInfant', '_75FCCInfant_flag',
       '_75FCCToddler', '_75FCCToddler_flag', '_75FCCPreschool',
       '_75FCCPreschool_flag'],
      dtype='object', length=227)
```

In [4]:
```
# print number of columns
num_columns = df.shape[1]
print(num_columns)
```

```
227
```

In [5]:
```
# print number of rows
num_rows = df.shape[0]
print(num_rows)
```

```
34567
```

In [6]:
```
# Find columns with the same value for all rows
constant_cols = [col for col in df.columns if df[col].nunique(dropna=False)
print("Columns with the same value in all rows:", constant_cols)
print("Number of such columns:", len(constant_cols))
```

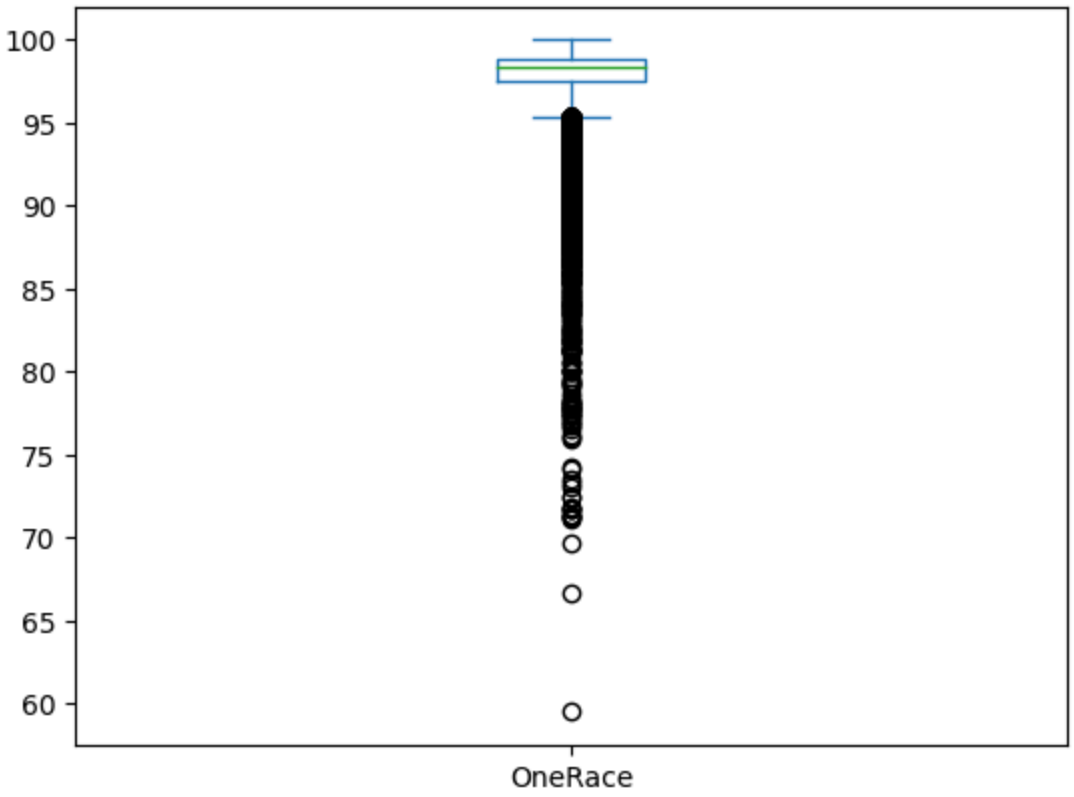
```
Columns with the same value in all rows: ['iEMP_M', 'iMEMP_M', 'iFEMP_M', 'i
EMP_Service', 'iMEMP_Service', 'iFEMP_Service', 'iEMP_Sales', 'iMEMP_Sales',
'iFEMP_Sales', 'iEMP_N', 'iMEMP_N', 'iFEMP_N', 'iEMP_P', 'iMEMP_P', 'iFEMP_
P']
Number of such columns: 15
```

In [7]:
```
# Calculate the average of the 'OneRace' column
average_one_race = df['OneRace'].mean()
print("Average of OneRace:", average_one_race)

# Create a boxplot of the 'OneRace' column
df['OneRace'].plot.box()

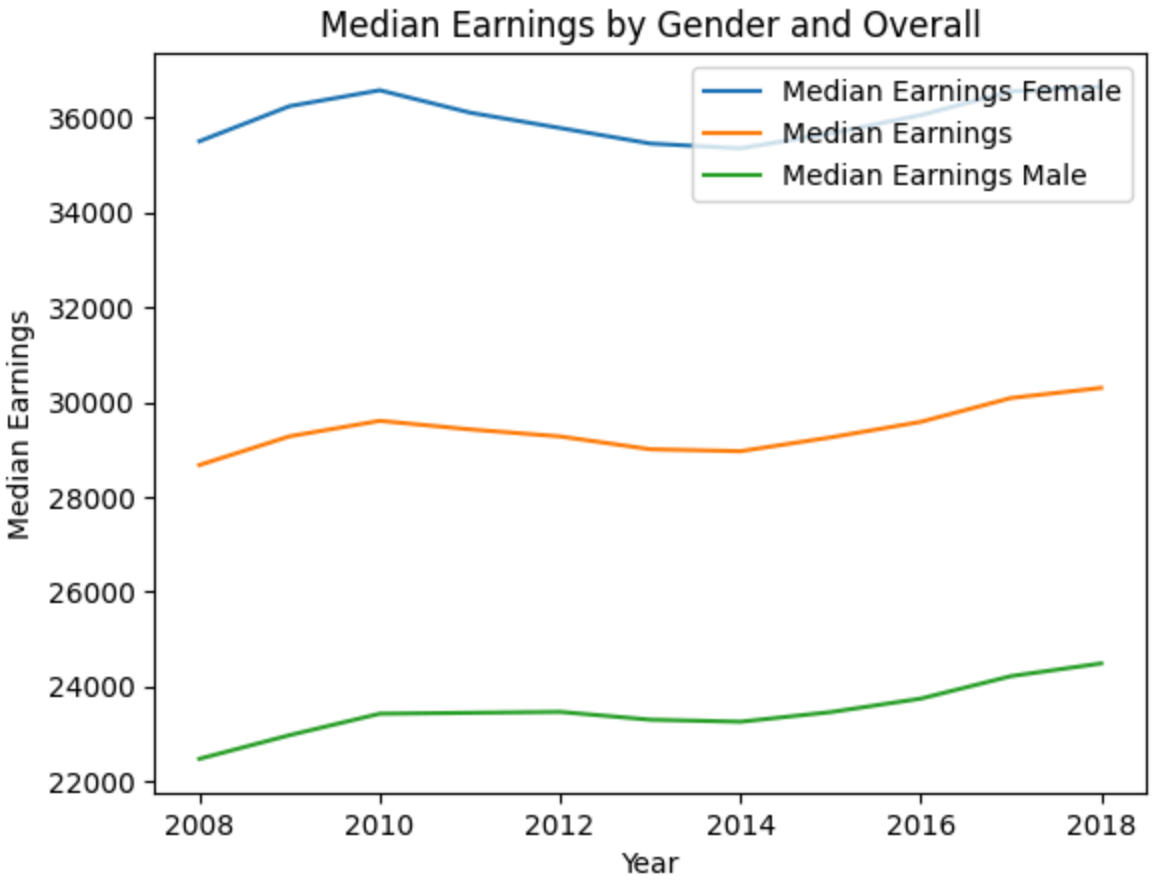
demographic_cols = ['OneRace_W','OneRace_B','OneRace_I','OneRace_A','OneRace
```

```
Average of OneRace: 97.92286284606705
```



In [8]:
```
# Plot Median Earnings Female, Median Earnings, and Median Earnings Male tog
df.groupby('StudyYear')[['MME_2018', 'ME_2018', 'FME_2018']].mean().plot(
    kind='line',
    title='Median Earnings by Gender and Overall',
    ylabel='Median Earnings',
    xlabel='Year'
).legend(['Median Earnings Female', 'Median Earnings', 'Median Earnings Male
```

Out[8]:
```
<matplotlib.legend.Legend at 0x319d29210>
```


Median Earnings by Gender and Overall
Median Earnings Female
Median Earnings
Median Earnings Male
Median Earnings
36000
34000
32000
30000
28000
26000
24000
22000
2008
2010
2012
2014
2016
2018
Year